



COURSE GUIDE

Introduction

Let's set up your development environment! You'll be using a few different tools in the Software Testing course, each of which you're likely to use in industry, so this is a great opportunity to get familiar with them all.

- **Git:** Git is a popular source control tool. It was written by Linus Torvalds (creator of Linux), and it's free and open source. You can read more about it at <https://www.git-scm.com>.
- **GitHub:** GitHub hosts Git repositories that multiple developers can access and collaborate on. Much of its functionality is free. Learn more at <https://www.github.com>.
- **Visual Studio Code:** VSCode is a free code editor published by Microsoft. Community-based plug-ins exist for most major computing languages. It also works well with Git and GitHub. Learn more at <https://code.visualstudio.com/>.
- **Python:** Python is a multi-paradigm programming language known for its ease of use and the number of community-developed libraries that support it. Curl up with more info at <https://www.python.org/>.
- **Node.js:** Node.js allows you to run JavaScript outside of a browser. It's an increasingly popular choice for web app backends. Read more about it at <https://nodejs.org/en/>.

Why aren't we learning to install these in a lesson? We want to focus the lessons on the how, what, and why of the software testing process. Installing your toolchain is an important part of that, but it's largely mechanical and best performed outside of the lessons.

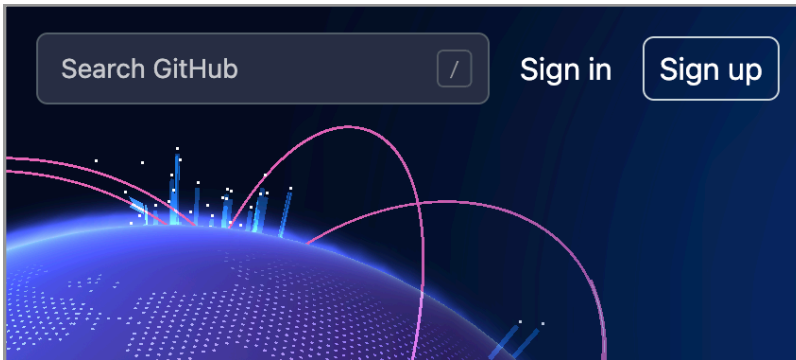
About this document. Unless otherwise noted, all Mac commands should be run in **Terminal**, and all Windows commands should be run in a **command line**. To avoid typos, all commands should be copied and pasted from this guide to your command interface.

Getting Started

Prerequisites

GitHub. Create and sign in to a GitHub account at www.github.com.

If you already have a GitHub account, sign in.

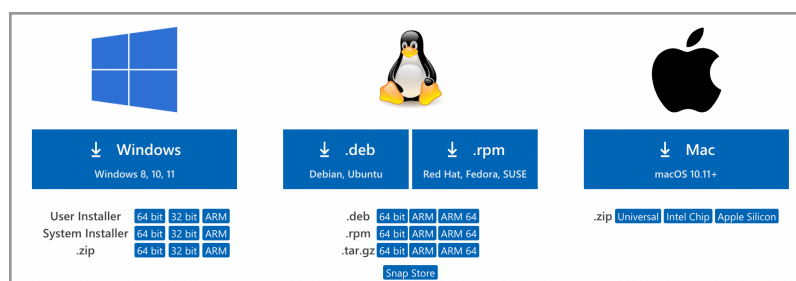


Navigate to <https://github.com/quanticedu/clean-test-course> and open the directory.

Familiarize yourself with the app by reading *README.md*.

	
	.github/workflows add backend coverage (#17)
	backend tax and total apis (#24)
	frontend Revert "test actions"
	readme-images Update readme (#14)
	.gitignore seed application
	README.md Update README.md

Visual Studio Code (VSCode). If you haven't installed VSCode, visit <https://code.visualstudio.com/download> and download the appropriate version. Then, create an account and sign in.



Installing Git

Command Line Interface

Check if Git is installed with the following command:

```
git --version
```

macOS. Visit <https://brew.sh> to install the Homebrew package manager. We recommend using the .pkg installer, but you can also use the command-line option on Homebrew's page.

Now run the following:

```
brew install git
```

Windows. Visit www.git-scm.com/download/win to download the appropriate interface for your machine's specifications.

If you don't have Git installed, proceed with the following steps.

To check if your machine has a 32- or 64-bit processor, open the Start menu, select System, and choose About. Find System Type under Device Specification, which displays processor info. Install the corresponding Git package.

Download for Windows

[Click here to download](#) the latest (2.37.2) 32-bit version of Git for Windows. This is the most recent maintained build. It was released 14 days ago, on 2022-08-12.

Other Git for Windows downloads

Standalone Installer

[32-bit Git for Windows Setup.](#)

[64-bit Git for Windows Setup.](#)



Updating/Installing Python and Node

Python

MAC. To check your version of Python, run the following:

```
python3 --version
```

Windows. To check your version of Python, run the following:

```
python --version
```

If you don't have Python installed or you're running Python v3.7 or lower, visit www.python.org/downloads/ to download the latest version.

Node.js

To check your version of Node on Mac or Windows, run the following:

```
node -v
```

If you don't have Node installed or are running Node v15 or lower, visit <https://nodejs.org/en/download/current/> to download the latest version.

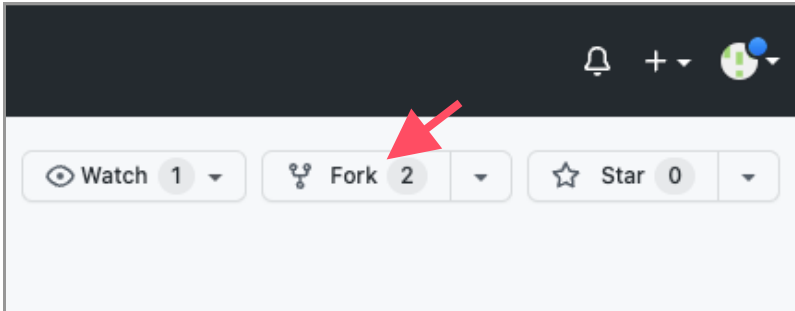
If you want to keep a legacy version of Python on your machine, check out the version manager located at <https://realpython.com/intro-to-pyenv/>.

Forking the Repository and Enabling Workflows

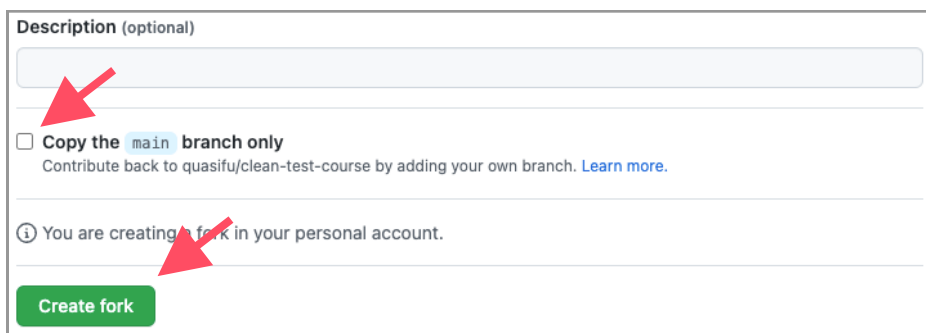
Forking A GitHub Repository

Navigate to <https://github.com/quanticedu/clean-test-course>.

Click “Fork” in the upper right corner.



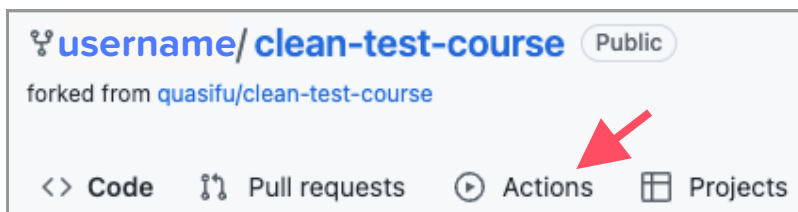
Uncheck the box stating “Copy the **main** branch only.” Leave all other settings as default, then click “Create fork.”



You'll be automatically placed in the forked repo.

Enabling Workflow Actions

Click “Actions” in the upper left corner.



Forking allows a developer to create a personal copy of a repository while maintaining file and versioning relationships with the original.

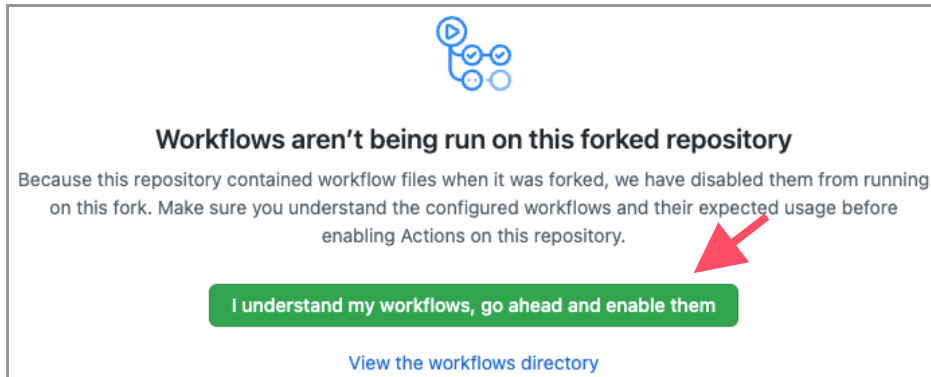
Branching allows a developer to create a branch within the original repository to develop new features or modifications without affecting the main code version.

Cloning allows a developer to create a copy of a repository on their local machine.

If you're not placed in the forked repo, you can access it from your GitHub homepage.

Forking the Repository and Enabling Workflows (continued)

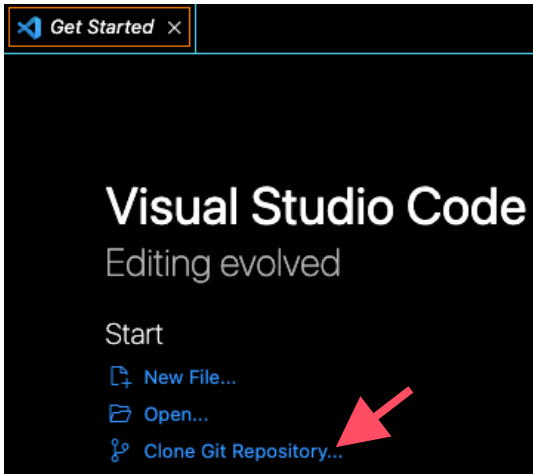
Click “I understand my workflows, go ahead and enable them.”



Workflows are disabled in new forks by default for any number of reasons. There may be secrets that need to be added before a workflow can run correctly, or there may be scheduled jobs that aren't desired in the fork.

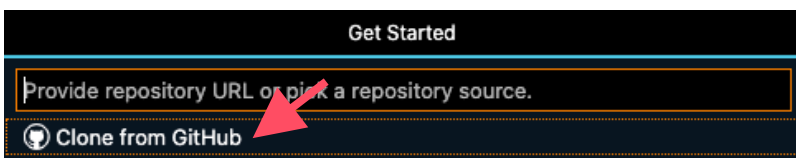
Cloning the Repository Locally in Visual Studio Code (VSCode)

On VSCode's Getting Started page, select "Clone Git Repository."



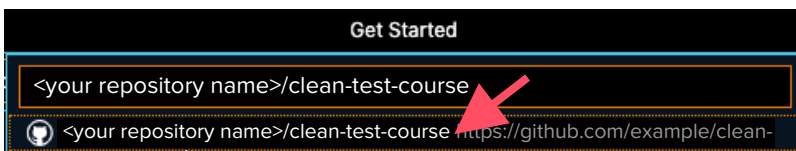
If you don't land at the Getting Started page when opening VSCode, select "Get Started" from the "Help" menu.

Select "Clone from GitHub." (Don't enter your repo in the box).



If this is the first time you've accessed GitHub from VSCode, you'll get a bunch of dialogs asking for permission for VSCode to access GitHub. Grant whatever permissions you're asked for.

Search "<your repository name>/clean-test-course" and select the option that appears.



When prompted, save the clone to your Desktop directory.

Errors

If you experience errors while cloning, you likely don't have an SSH key. Use the guide located at <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent> to generate a new SSH key.

Afterward, add it to your Github account with the instructions at <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account>.

Navigating Repository Branches Using VSCode

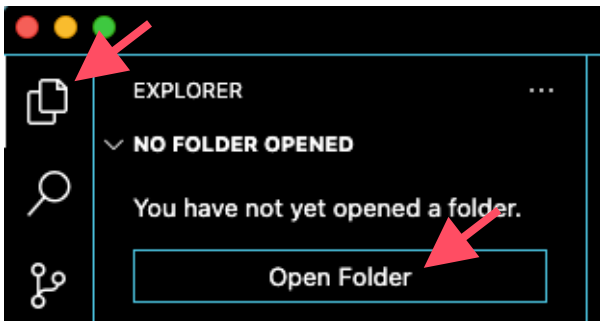
At the start of most lessons in this course, you'll be directed to open a corresponding branch of code in *clean-test-course*. The branches we're concerned with are named in one of two ways:

`origin/lesson-X-complete`

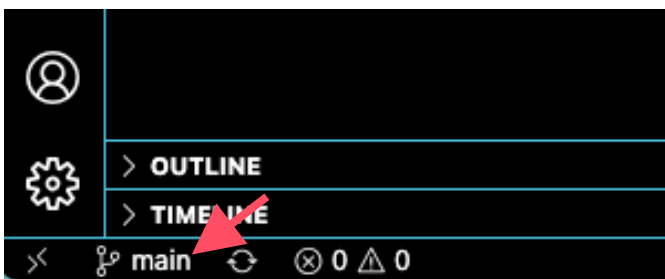
`origin/lesson-X`

You won't use the `complete` branches; they're here to help the application run. They can be used as a reference if you have trouble in your branch.

In VSCode, click the Explorer icon in the top left corner. If you see "You have not opened a folder," select "Open Folder." Navigate to your Desktop and open *clean-test-course*.

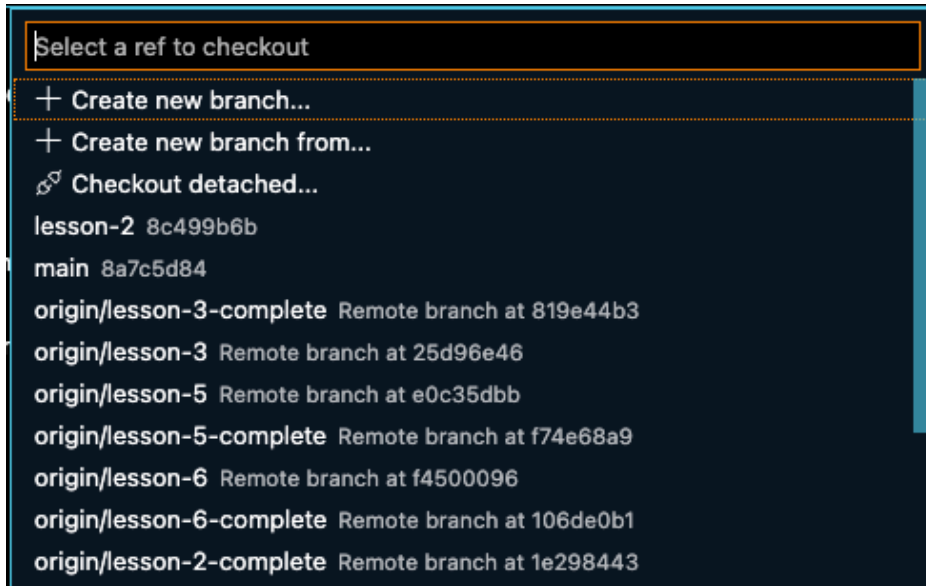


In the bottom left corner, click on "main."



Navigating Repository Branches (continued)

In the dropdown window, select the branch that pairs with your current lesson.



Select **origin/lesson-2** if you're taking lesson 2, for example. You'll be told exactly which branch to open early in applicable lessons.

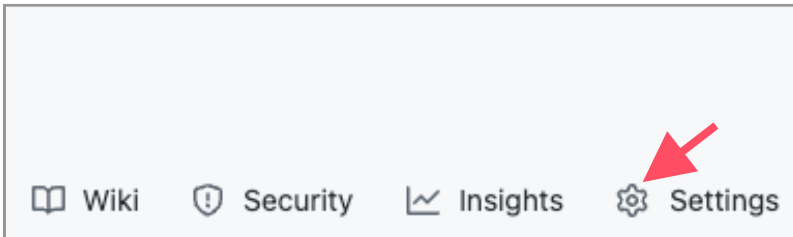
These steps can be used to switch between branches at any time. Once you've selected a branch, any file edits made will only be applied in that branch. Other branches will remain in their current state.

Unforking the Repository

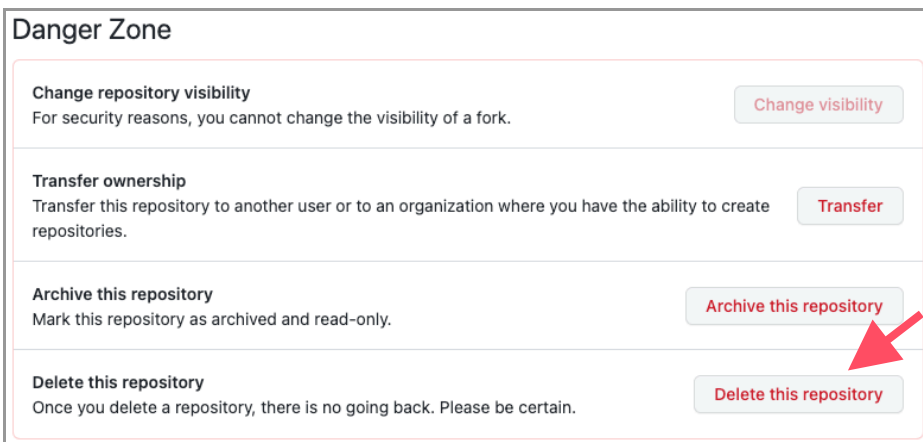
UPON COURSE COMPLETION ONLY

Warning. This action is permanent, so ensure you've completed every lesson and the Smart Case before proceeding.

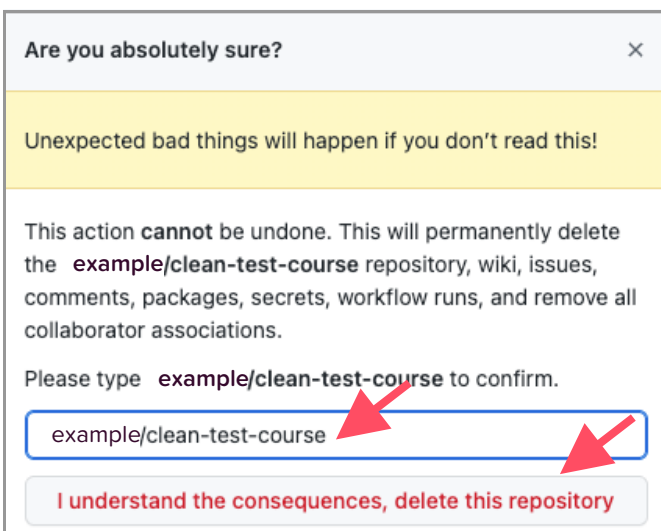
In your forked repository, click “Settings” near the top of the screen.



Scroll to the Danger Zone and select “Delete this repository.”



Enter the repository name and select “I understand the consequences, delete this repository”.



Make absolutely sure you're ready to delete the repository. There's no going back!